



Vigil VMS
User Manual
Version 1.0

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1. Introduction

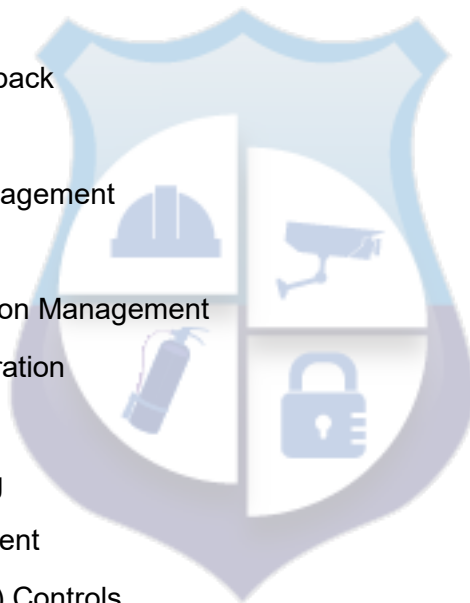
Vigil VMS is an intelligent Video Management System (VMS) designed to provide centralized surveillance monitoring, recording, playback, AI-powered analytics, and event management through a unified platform.

The system supports IP Cameras, NVRs, RTSP Streams, AI-assisted analytics, cloud backup integration, multi-site monitoring, and role-based access control for commercial and enterprise surveillance deployments.

2. System Overview

Vigil VMS provides:

- Camera Discovery and Onboarding
- Live Monitoring
- Recording and Playback
- AI Analytics
- Event and Alert Management
- Snapshot Services
- Storage and Retention Management
- Cloud Backup Integration
- User Administration
- Multi-Site Monitoring
- Licensing Management
- Pan-Tilt-Zoom (PTZ) Controls.
- Import/Export of Video Clips.



3. Key Features

Surveillance Management

- Multi-camera monitoring
- Camera grouping
- Multi-site management
- Recording and playback
- Snapshot generation

AI Analytics

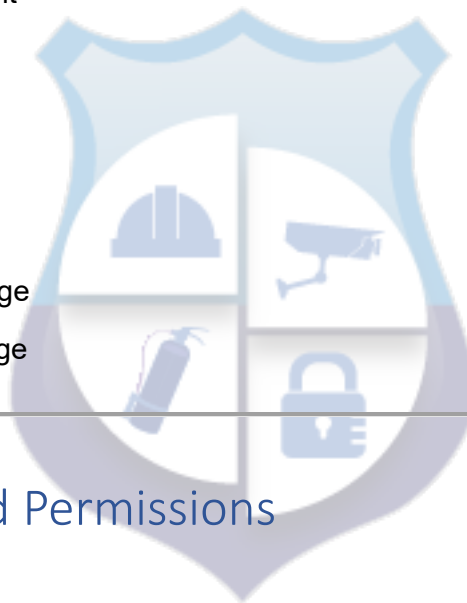
- Person Detection
- Vehicle Detection
- Animal Detection
- Fire Detection
- Zone Intrusion Detection
- Intelligent Alerting

Administration

- Role-Based Access Control (RBAC)
- User Management
- License Management
- Health Monitoring

Storage

- Local Storage
- NAS Storage
- Cloud Backup Storage
- Local + Cloud Storage



4. User Roles and Permissions

Administrator

Administrators can:

- Manage cameras
- Configure AI analytics
- Configure storage
- Manage users
- Configure cloud backup
- Manage licenses
- Access all system settings
- Bulk Import/Export

Operator

Operators can:

- View live feeds
- Review recordings
- Investigate alerts
- Export recordings
- Review AI events

Viewer

Viewers can:

- View assigned cameras
- Review permitted recordings
- View Snapshots
- Get AI Alerts

5. Getting Started

Minimum Requirements

- 8 GB RAM Minimum
- 16 GB RAM Recommended
- Stable Network Connection
- Supported IP Cameras, NVRs, or RTSP Streams

First Login

1. Launch Vigil VMS.
2. Create Username and Password.
3. Click Signup.
4. Then Login with same Credentials.

6. Login and Authentication

Vigil VMS uses secure authentication and Role-Based Access Control (RBAC).

Each user is granted permissions based on their assigned role.

Unauthorized operations are automatically restricted.

7. Dashboard Overview

The Dashboard provides:

- Live View Panel
- Event & Alert Panel
- Camera Status Overview
- Storage Status
- AI Analytics Status
- System Health Indicators
- Site Selection Menu

8. Camera Management

Adding a Camera

1. Navigate to Camera Management.
2. Click Add Camera.
3. Enter:
 - Camera Name
 - IP Address
 - RTSP URL
 - Username
 - Password
4. Click Test Connection.
5. Click Save.

Automatic Discovery

1. Open Camera Discovery.
2. Click Scan Network.
3. Select detected devices.
4. Click Import.



Adding Cameras in Bulk through NVR

1. Open Camera Discovery.
 2. Click Bulk Import.
 3. Enable Add through NVR.
 4. Add NVR IP address.
 5. Enter number of channels, port, username and password.
 6. Enable "Skip Duplicates" to avoid adding same cameras again.
 7. You can upload cameras in bulk by creating a CSV/JSON file with credentials of the cameras to be added and then uploading it in the App.
-

9. Live Monitoring

1. Open Live View.
2. Select a layout.
3. Assign cameras to view slots.
4. Double-click a stream for Full Screen.

Supported layouts: from 1x1 upto 7x7 grid size.

10. Recording Management

Supported Modes:

- Manual Recording
- Scheduled Recording
- Continuous Recording
- Priority based Recording

Manual Recording

1. Open Live/grid View.
2. Select Camera.
3. Click Record.

The recording indicator confirms active recording.

11. Playback and Investigation

1. Open Playback.
2. Select Camera.
3. Choose Date and Time.
4. Review footage using Timeline Controls.
5. Add bookmarks and events if required.

Exporting Recordings

1. Select Start Time.
2. Select End Time.
3. Click Export.
4. Select destination folder.

12. AI Analytics Configuration

AI Analytics can be enabled individually for licensed cameras.

Supported Analytics

- Person Detection
- Vehicle Detection
- Animal Detection
- Fire Detection
- Zone Intrusion Detection

Enable Analytics

1. Open Camera Management.
2. Select Camera.
3. Open AI Analytics Settings.
4. Enable required analytics modules.
5. Click Save.

If AI licenses are unavailable, activation will be blocked until additional AI licenses are assigned.

13. Event and Alert Management

Detected events automatically appear in the Alerts Panel.

Each event contains:

- Event Type
- Camera Source
- Timestamp
- Event Snapshot

Operators can:

- Review Alerts
- Acknowledge Events
- Open Playback
- Export Evidence through playback.

14. Snapshot Management

Snapshots are automatically generated during AI events.

Users can:

- View Snapshots
- Download Snapshots

15. Storage Management

Supported Storage Types:

Local Storage

Primary recording storage.

NAS Storage

Long-term archival storage.

Cloud Backup Storage

Remote backup and disaster recovery storage.

Administrators can define retention and archival policies.

16. Cloud Backup Storage Configuration

Overview

Vigil VMS supports Amazon S3 and S3-compatible storage providers for cloud archival of surveillance recordings.

PART 0 — Create AWS Account (VERY IMPORTANT FIRST)

1. Go to AWS

Open:

<https://aws.amazon.com>

Click **Create an AWS Account**

Fill:

- Email → use your permanent email (not college email)
 - Password → strong password
 - Account name → VigilVMS (or anything)
-

2. Payment Method

AWS requires a card (₹2 temporary charge → refunded)

Even free tier requires card

You will NOT be charged if we follow settings correctly.

3. Verification

- Phone OTP
- Choose **Personal account**
- Choose **Basic Support (Free)**

Done → wait ~2 minutes

PART 1 — SECURE THE ACCOUNT (MOST PEOPLE SKIP → THEN GET HACKED)

After login:

Top right → your name → **Security credentials**

Enable MFA (Mandatory)

Click → Assign MFA device

Install on phone:
Google Authenticator / Microsoft Authenticator
Scan QR → enter code → Finish

DO NOT USE ROOT USER AGAIN

We now create an admin user.

Go to:
AWS Console → search **IAM**

PART 2 — CREATE ADMIN USER (FOR DAILY USE)

IAM → Users → Create user

User name:

admin-user

Tick:

☒ Provide user access to AWS Management Console

Set password → custom password

Next → Permissions:

Choose:

☒ Attach policies directly

Search:

AdministratorAccess

Select it → Next → Create user



Login as admin user

Sign out

Now login using:

<https://signin.aws.amazon.com/console>

(You will see this URL in IAM dashboard)

Username: admin-user

Never login with root again.

PART 3 — CREATE S3 BUCKET (STORAGE FOR VIDEOS)

Go to Services → **S3**

Click **Create bucket**

Step-by-step EXACT SETTINGS

Bucket name

Must be globally unique:

vigil-recordings-mrunali-01

Rules:

- lowercase only
 - no spaces
 - no underscore
-

Region (IMPORTANT FOR SPEED)

Choose closest to India:

Asia Pacific (Mumbai) ap-south-1

⚠ Remember this:

ap-south-1

Object Ownership

Keep default:

ACLs disabled (recommended)

Block Public Access

KEEP ALL CHECKED ☒

Very important for security

Bucket Versioning

Disable (saves money)

Encryption

Enable:

☒ Server-side encryption (SSE-S3)

Click **Create bucket**

✓ Bucket created

PART 4 — CREATE POLICY (PERMISSION ONLY FOR THIS BUCKET)

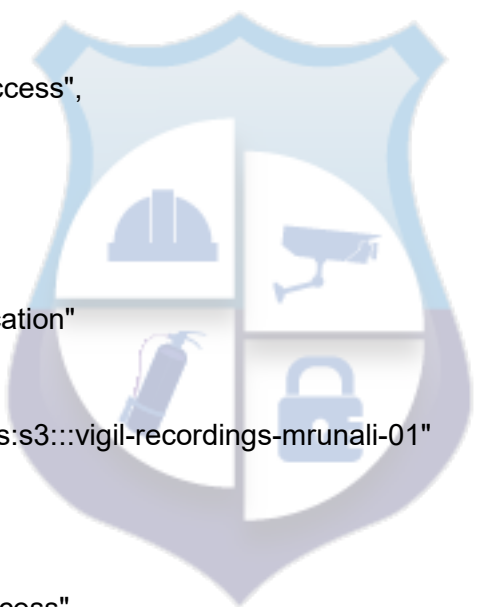
Go to:

IAM → Policies → Create policy → JSON tab

Paste this:

(Replace bucket name)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "VigilBucketAccess",
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket",
        "s3:GetBucketLocation"
      ],
      "Resource": "arn:aws:s3:::vigil-recordings-mrunali-01"
    },
    {
      "Sid": "VigilObjectAccess",
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:GetObject",
        "s3:DeleteObject",
        "s3:AbortMultipartUpload"
      ],
      "Resource": "arn:aws:s3:::vigil-recordings-mrunali-01/*"
    }
  ]
}
```



}

Next

Policy name:

VigilS3AccessPolicy

Create policy

PART 5 — CREATE PROGRAMMATIC USER (USED BY VIGIL)

IAM → Users → Create user

Name:

vigil-cloud-user

Next → Permissions

Attach:

VigilS3AccessPolicy

Create user

PART 6 — CREATE ACCESS KEYS (MOST IMPORTANT STEP)

Open user → Security credentials → Access keys → Create access key

Choose:

Application running outside AWS

Create

You will get:

Access Key ID: AKIAXXXXXX

Secret Access Key: XXXXX

⚠ SAVE THIS NOW (never shown again)

PART 7 — CONFIGURE VIGIL VMS

Open Vigil dashboard:

Settings → Cloud Configuration

Fill EXACTLY:

Field	Value
Provider	S3 Compatible
Bucket Name	vigil-recordings-mrunali-01
Endpoint URL	https://s3.ap-south-1.amazonaws.com
Region	ap-south-1
Path Prefix	recordings/
Access Key	AKIA...
Secret Key	your secret

Click **Save**

PART 8 — TEST CONNECTION

If working → status becomes:

HEALTHY

If error → check common mistakes below

MOST COMMON ERRORS (95% PEOPLE GET THESE)

Error: Access Denied

Bucket name mismatch OR policy wrong

Error: Could not connect to endpoint

Wrong region endpoint

Correct:

<https://s3.ap-south-1.amazonaws.com>

NOT:

<https://s3.amazonaws.com>

Error: Signature mismatch

Wrong secret key OR extra spaces copied

Error: Works but no upload

Path prefix missing slash

Correct:

recordings/

Wrong:

recordings

OPTIONAL — COST CONTROL (VERY IMPORTANT)

Go to:

Billing → Budgets → Create budget

Monthly:

₹200

Alert at:

₹50

₹100

₹150

This prevents surprise bills.

FINAL RESULT

Your videos will automatically be stored remotely even if device is stolen.

17. Licensing and Subscription Management

Licensing Overview

Vigil VMS operates under a licensed deployment model.

Licenses are issued and managed by the Vigil Support Team.

For:

- New Licenses
- Renewals
- Upgrades
- Capacity Expansion

Contact the Vigil Support Team.

Available Plans

Basic Plan :

Supports up to 4 Cameras.

Professional Plan :

Supports up to 16 Cameras.

Enterprise Plan :

Supports 1,000+ Cameras.



License Activation

1. Login as Administrator.
2. Navigate to:

Administration → Licensing

3. Click Activate License.
4. Enter License Key.
5. Click Validate.
6. Click Activate.

The system will display:

- Plan Name
- Camera Capacity
- License Status

AI Analytics Licensing

AI Analytics is licensed separately from the base VMS subscription.

Supported Features:

- Person Detection
- Vehicle Detection
- Animal Detection
- Fire Detection
- Zone Intrusion Detection
- Intelligent Event Generation
- AI-Based Alerting

AI Licensing Model

AI licenses are allocated per camera.

Example:

Professional Plan (16 Cameras)

If AI Analytics is required on only 4 cameras, then a customised AI license for just 4 cameras is created by Vigil Support Team and can be used on any 4 of your cameras.

The remaining cameras continue operating with standard VMS functionality.

Managing AI Licenses

Navigate to:

Administration → Licensing

Administrators can view:

- Total AI Licenses
- Assigned AI Licenses
- Available AI Licenses

18. Multi-Site Monitoring

Features include:

- Site Grouping
- Camera Grouping
- Centralized Monitoring

- Unified Alert Management
 - Cross-Site Operations
 - Video Wall Integration and Management
-

19. System Health Monitoring

Monitor:

- Camera Connectivity
 - Recording Status
 - Storage Usage
 - AI Service Status
 - Cloud Connectivity
 - System Services
 - NAS
-

20. Backup and Recovery

Recommended Practices:

- Configure Cloud Backup
 - Test Recovery Procedures
 - Review Retention Policies
 - Verify Archive Availability
-

21. Troubleshooting

Camera Offline

Possible Causes:

- Network Issues
- Invalid RTSP URL
- Incorrect Credentials

Cloud Access Denied

Possible Causes:

- Invalid Bucket Permissions

- Incorrect IAM Policy

AI Analytics Not Working

Possible Causes:

- Analytics Disabled
- No AI License Assigned
- AI Service Offline

Upload Failure

Possible Causes:

- Incorrect Cloud Configuration
- Invalid Credentials
- Network Issues

22. Best Practices

- Use strong passwords.
- Configure cloud backups for critical deployments.
- Review storage utilization regularly.
- Assign AI Analytics only where required.
- Periodically review user permissions.

23. Support

For technical support, licensing assistance, AI Analytics licensing, cloud storage setup, deployment guidance, or troubleshooting assistance, contact the Vigil Support Team.